

Introduction to Artificial Neural Networks (ANN),

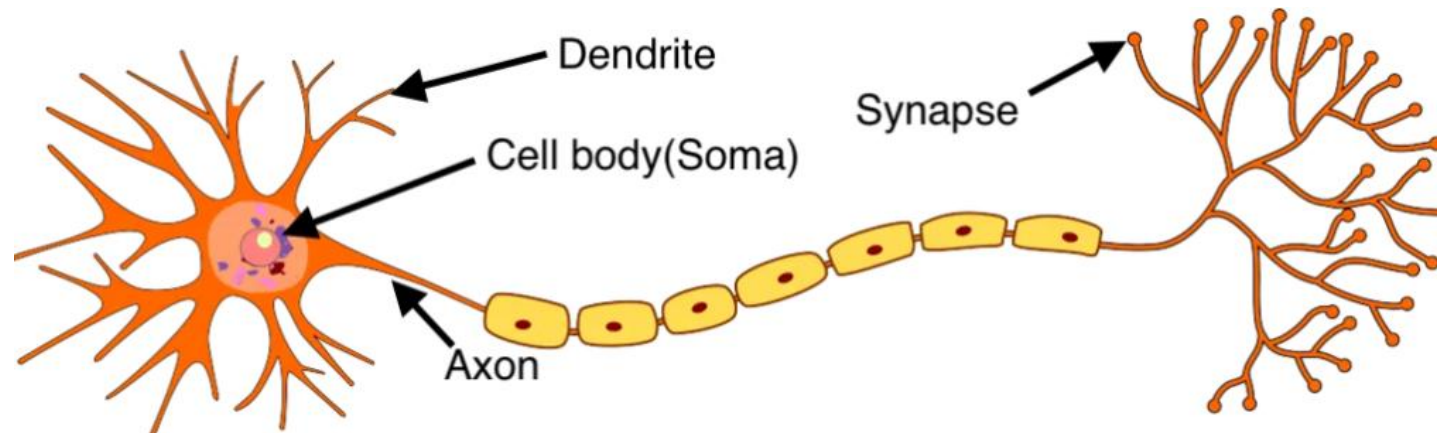
Neural Network

- **Neural network:** *information processing paradigm inspired by biological nervous systems, such as our brain.*
- Structure: large number of highly interconnected processing elements (*neurons*) working together
- *Like people, they learn from experience (by example) It is configured for special application such as pattern recognition and data classification through a learning process.*

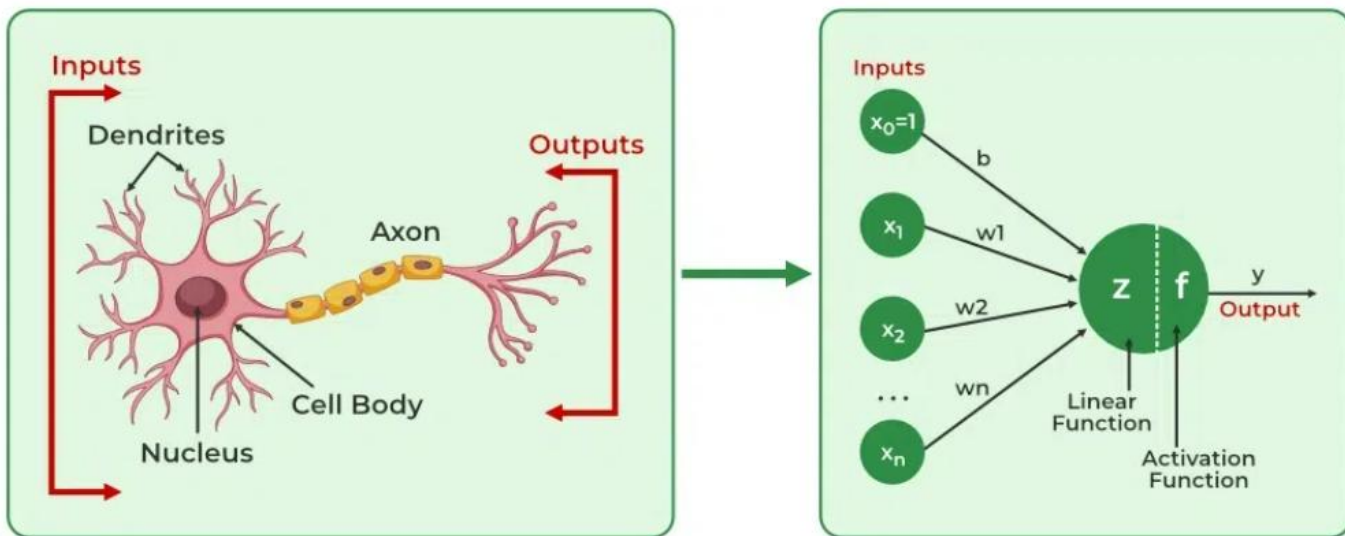
Artificial Neurons

• ANN is an information processing system that has certain performance characteristics in common with biological nets.

1. *The processing element receives many signals.*
2. *Signals may be modified by a weight at the receiving synapse.*
3. *The processing element sums the weighted inputs.*
4. *Under appropriate circumstances (sufficient input), the neuron transmits a single output.*
5. *The output from a particular neuron may go to many other neurons.*

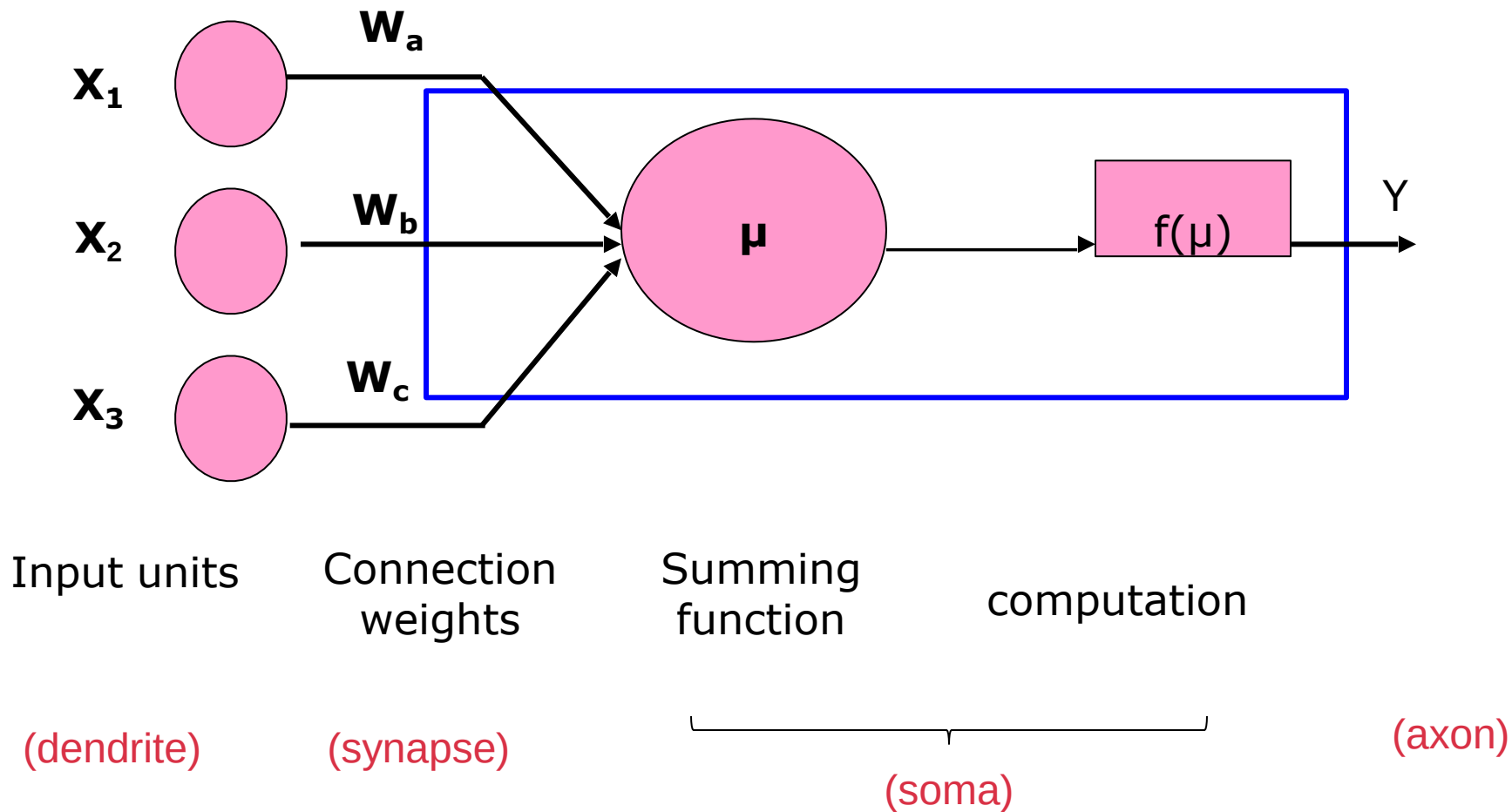


Terminology Relation Between Biological And Artificial Neuron



Biological Neuron	Artificial Neuron
Cell	Neuron
Dendrites	Weights or interconnections
Soma	Net input
Axon	Output

Model Of A Neuron



Model Of A Neuron

- A neural net consists of a large number of simple processing elements called neurons, units, cells or nodes.
- Each neuron is connected to other neurons by means of directed communication links, each with associated weight.
- The weight represent information being used by the net to solve a problem.

Model Of A Neuron

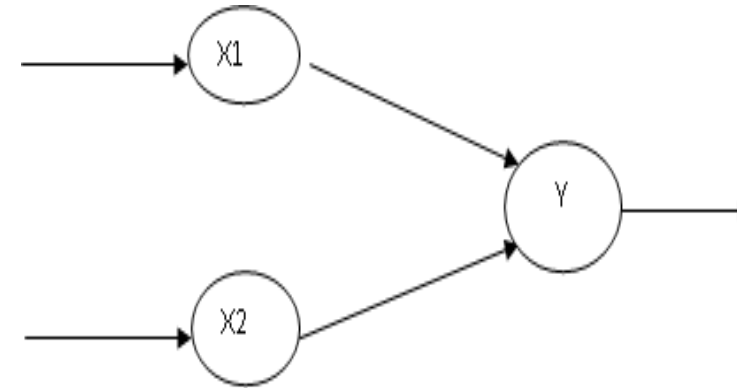
- Each neuron has an internal state, called its activation or activity level, which is a function of the inputs it has received. Typically, a neuron sends its activation as a signal to several other neurons.
- It is important to note that a neuron can send only one signal at a time, although that signal is broadcast to several other neurons.

Application Scope of Neural Networks

- Air traffic control
- Animal behavior
- Appraisal and valuation of property, etc.,
- Betting on horse races, stock markets
- Criminal sentencing
- Complex physical and chemical process
- Data mining, cleaning and validation
- Direct mail advertisers
- Echo patterns
- Economic modeling
- Employee hiring
- Expert consultants
- Fraud detection
- Hand writing and typewriting
- Lake water levels
- Machinery controls
- Medical diagnosis
- Music composition
- Photos and finger prints
- Recipes and chemical formulation
- Traffic flows
- Weather prediction

Basic Operation of a Neural Net

- X1 and X2 – input neurons.
- Y- output neuron
- Weighted interconnection links- W1 and W2.
- Net input calculation is :



$$Y_{in} = x_1w_1 + x_2w_2 + \dots + x_nw_n$$

$$\sum_{i=1}^n x_iw_i$$

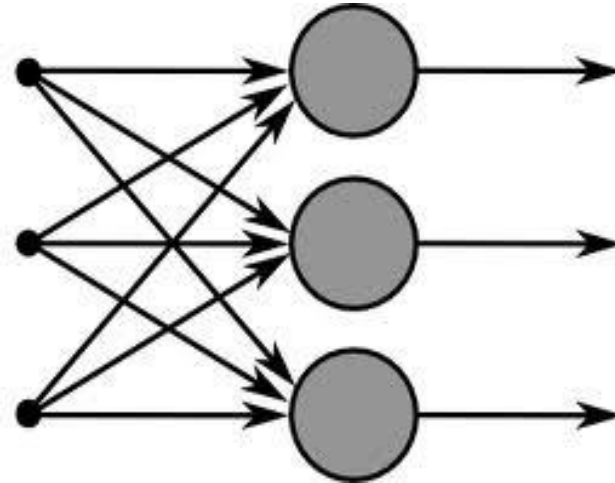
- Output is : $y = f(Y_{in})$

Basic models of ANN

- Models are based on three entities
 - *The model's synaptic interconnections.*
 - *The training or learning rules adopted for updating and adjusting the connection weights.*
 - *Their activation functions*
- The arrangement of neurons to form layers and the connection pattern formed within and between layers is called the **network architecture**.
- Five types:
 - *Single layer feed forward network*
 - *Multilayer feed-forward network*
 - *Single node with its own feedback*
 - *Single-layer recurrent network*
 - *Multilayer recurrent network*

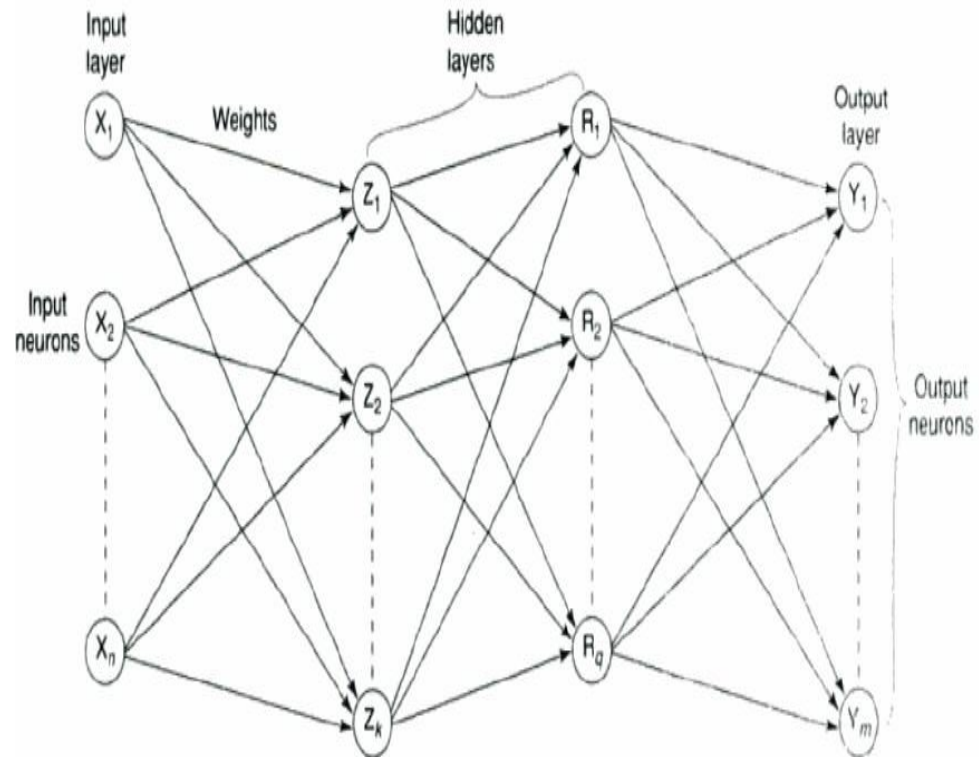
Single layer Feed- Forward Network

- Layer is formed by taking processing elements and combining it with other processing elements.
- Input and output are linked with each other
- Inputs are connected to the processing nodes with various weights, resulting in series of outputs one per node.



Multilayer feed-forward network

- Formed by the interconnection of several layers.
- Input layer receives input and buffers input signal.
- Output layer generated output.
- Layer between input and output is called *hidden layer*.
- Hidden layer is internal to the network.
- Zero to several hidden layers in a network.
- More the hidden layer, more is the complexity of network, but efficient output is produced.



Feed back network

- If no neuron in the output layer is an input to a node in the same layer / proceeding layer – **feed forward network**.
- If outputs are directed back as input to the processing elements in the same layer/proceeding layer – **feedback network**.
- If the output are directed back to the input of the same layer then it is **lateral feedback**.
- **Recurrent networks** are networks with feedback networks with closed loop.
- Fig 2.8 (A) –simple recurrent neural network having a single neuron with feedback to itself.
- Fig 2.9 – single layer network with feedback from output can be directed to processing element itself or to other processing element/both.

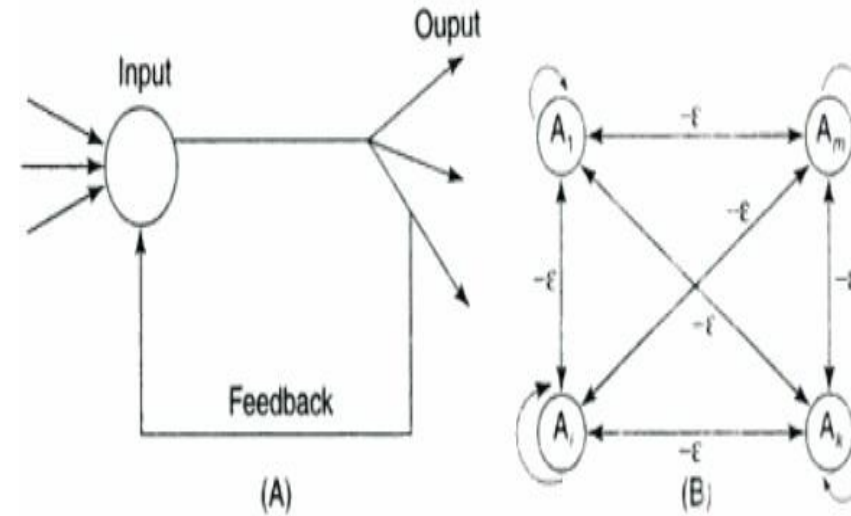


Figure 2-8 (A) Single node with own feedback. (B) Competitive nets.

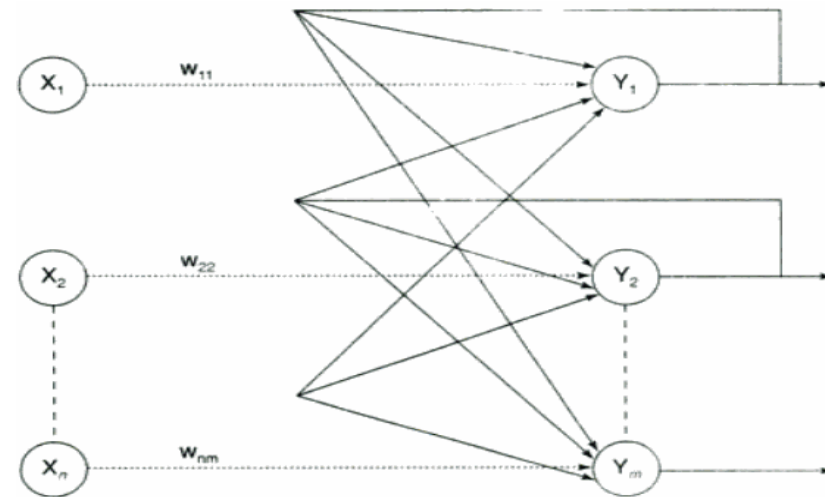
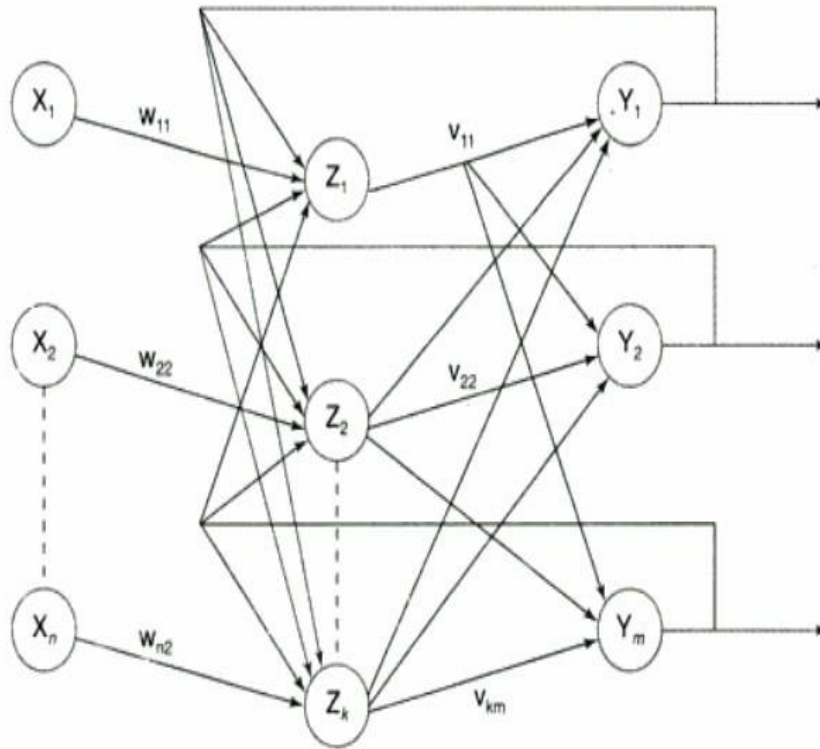


Figure 2-9 Single-layer recurrent network.



- Processing element output can be directed back to the nodes in the preceding layer, forming a ***multilayer recurrent network***.
- Processing element output can be directed to processing element itself or to other processing element in the same layer.

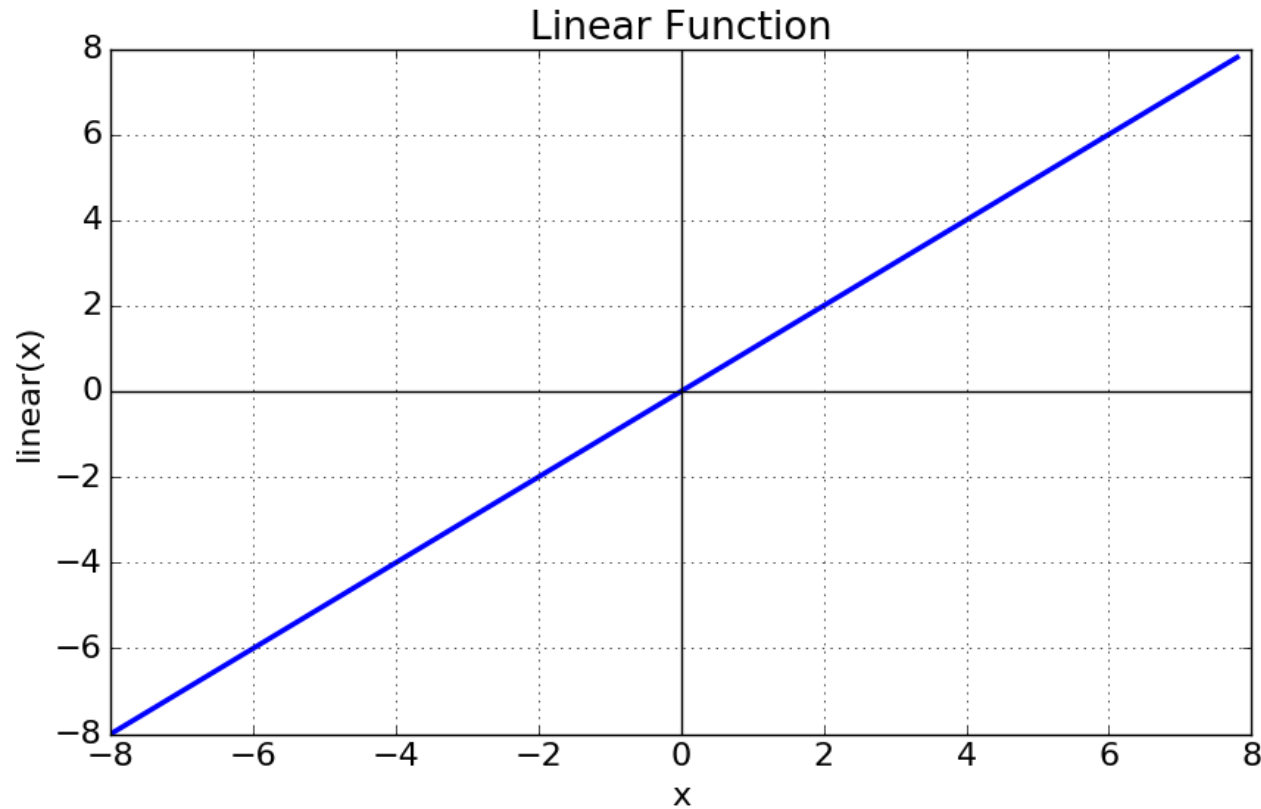
Activation Function

Activation function is used to determine the output of neural network like yes or no. It maps the resulting values in between 0 to 1 or -1 to 1 etc. (depending upon the function).

Types:

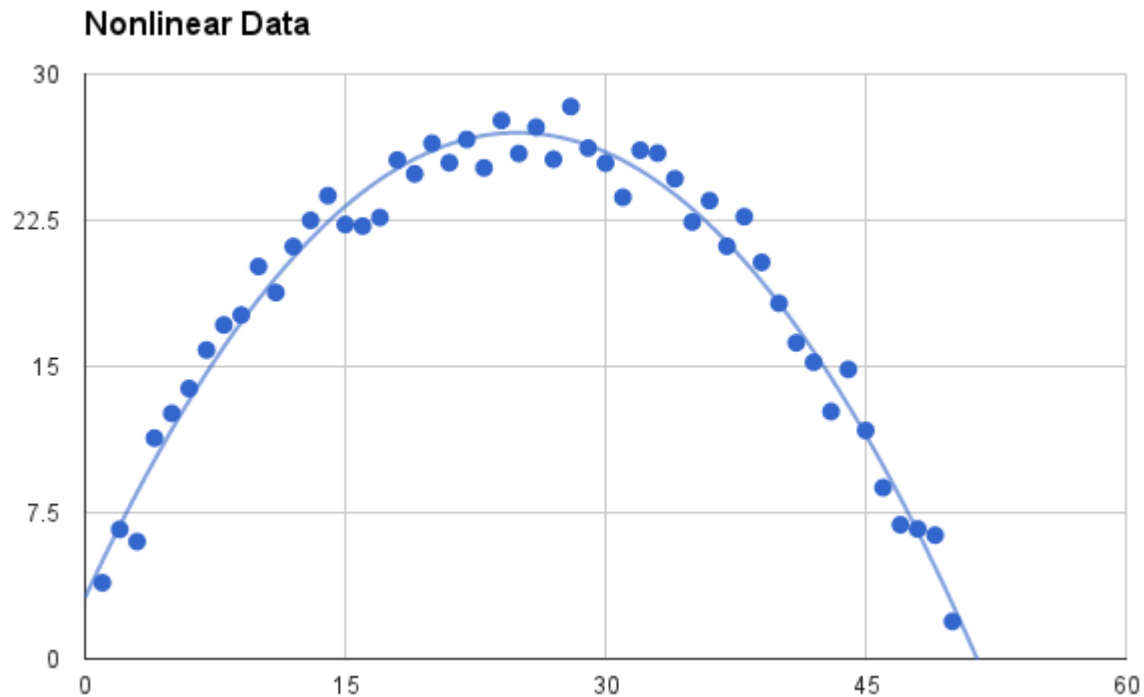
- ◎ Linear Activation Function
- ◎ Non-linear Activation Functions

Linear or Identity Activation Function



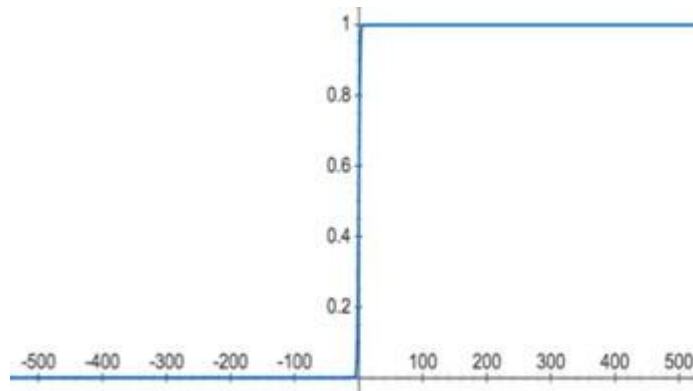
- **Equation :** $f(x) = x$
- **Range :** $-\infty$ to $+\infty$
- It doesn't help with the complexity or various parameters of usual data that is fed to the neural networks.

Non-linear Activation Function



- **Derivative or Differential:** Change in y-axis w.r.t. change in x-axis. It is also known as slope.
- **Monotonic function:** A function which is either entirely non-increasing or non-decreasing.

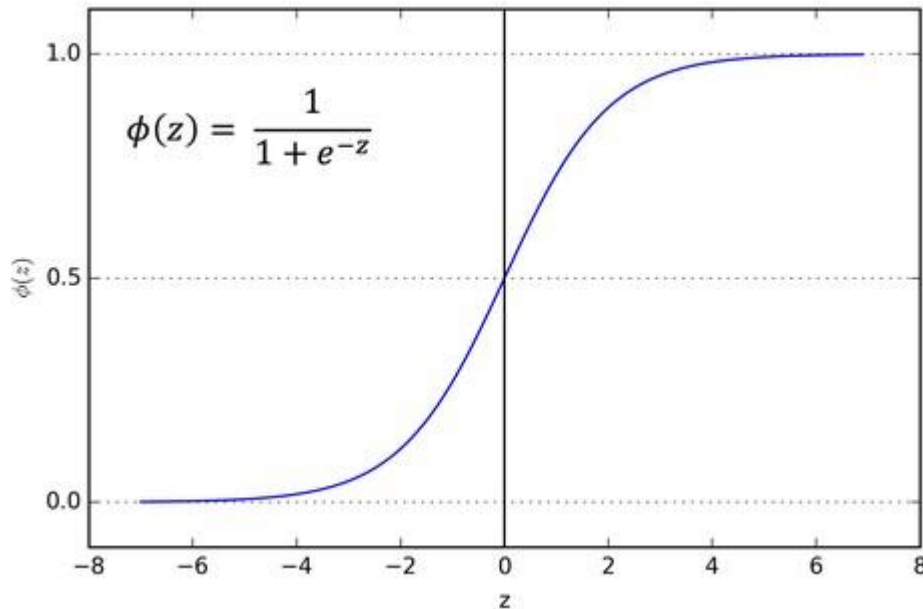
Binary Step Function



$$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{For } x \geq 0 \end{cases}$$

- Useful for binary classifier
- Not useful for multi-class classification problem.
- Derivative is 0 which means backpropagation is not possible due to the vanishing gradients problem.

Sigmoid or Logistic Activation Function

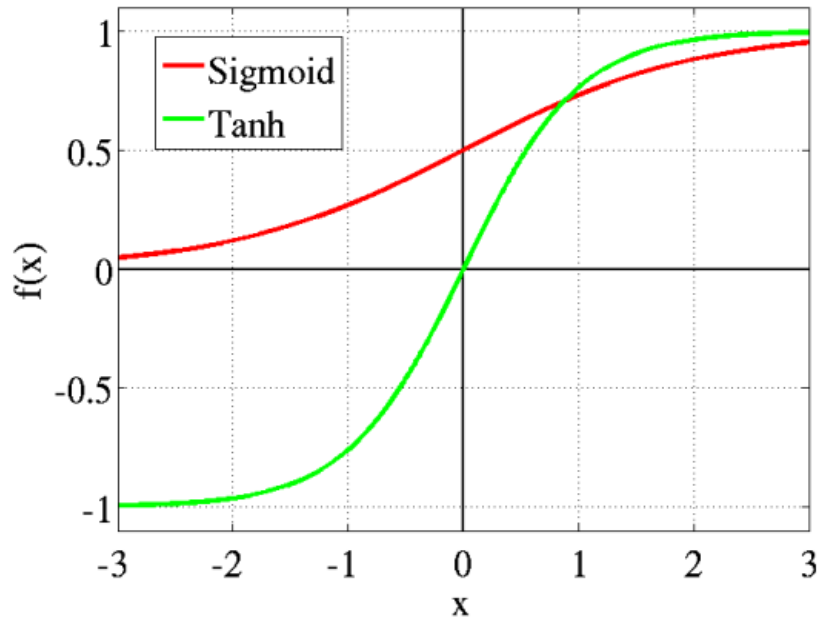


- **Equation** $\phi(z) = \frac{1}{1 + e^{-z}}$
- **Range** : 0 to 1
- It is especially used for models where we have to **predict the probability** as an output.
- **Differentiable** ○ **Monotonic**

The logistic sigmoid function can cause a neural network to get stuck at the training time.

The **softmax function** is a more generalized logistic activation function which is used for multiclass classification.

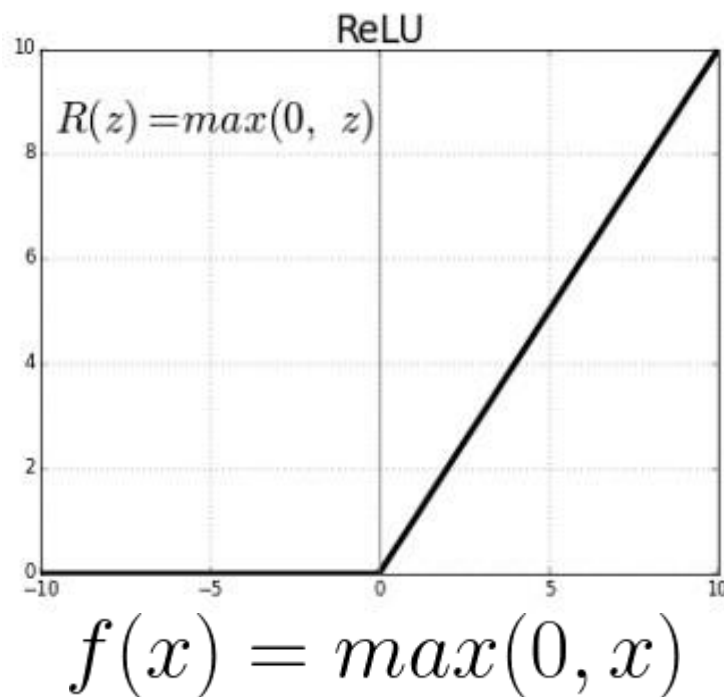
Tanh or hyperbolic tangent Activation Function



$$f(x) = \tanh(x) = \frac{2}{(1 + e^{-2x})} - 1$$

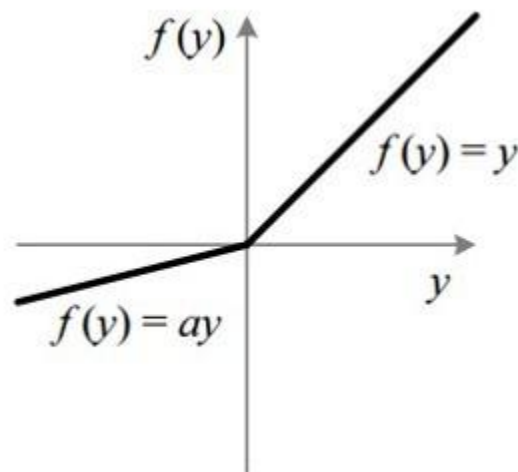
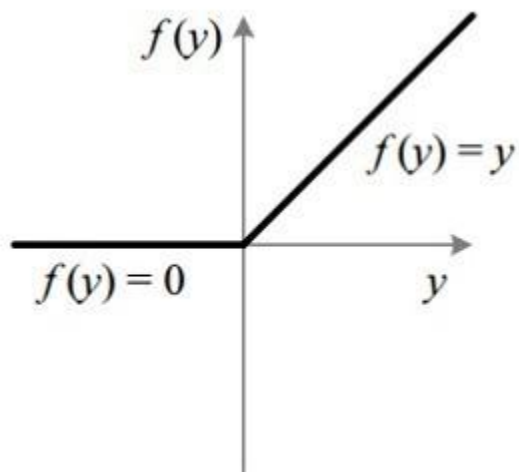
- **'S' shaped like sigmoid**
- **Range : -1 to 1**
- Negative inputs will be mapped strongly negative and the zero inputs will be mapped near zero in the tanh graph.
- **Differentiable**
- **Monotonic**
- Used for classification between two classes.
- *Both tanh and logistic sigmoid activation functions are used in feed-forward nets.*

ReLU (Rectified Linear Unit) Activation Function



- ReLU is half rectified (from bottom).
- **Range:** [0 to infinity)
- The function and its derivative **both are monotonic.**
- Widely used in CNN.
- Easy to compute
- Does not saturate
- Does not cause the Vanishing Gradient Problem.
- Not zero centred.
- “**dying ReLU**” problem

Leaky ReLU



- › Solves the dying ReLU problem
- › Usually, the value of **a** is 0.01 or so.
- › When **a is not 0.01** then it is called **Randomized ReLU**.
- › **Range:**
- **Monotonic**



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